SINGAPORE DIMENSIONS • GRADE 2A • CHAPTER 1

Numbers to 1,000: Hundreds, Tens & Ones

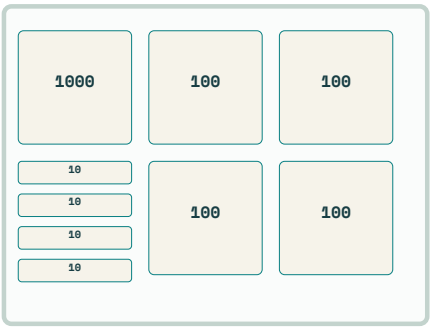
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.2.NBT.A.1

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Place Value Base Ten

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
3 flats (300) + 4 rods (40) + 5 units (5) form 345 on the desk.

Sketch Model on Paper:
Sketch expanded notation blocks: 300 + 40 + 5.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE: 123 Base-10 Arithmetic

1.	4	Step 1	Execute [4]
2.	0	Step 2	Execute [0]
3.	0	Step 3	Execute [0]
4.	enter	Step 4	Execute [enter]

EXPECTED LCD X: 478

1 **Combine 200 + 50 + 6.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Combine 500 + 30 + 0.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Combine 600 + 0 + 9.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **What is 100 more than 450?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Chapter 1

Numbers to 1,000: Hundreds, Tens & Ones • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Place Value Studio (Flats, Rods, Units)

CAD Blueprint: place_value_base_ten

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/lib/place_value_base_ten.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

1000 Blue Cube, 100 Yellow Flats, 10 Green Rods (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Combine 200 + 50 + 6.** Standard: CCSS.MATH.2.NBT.A.1	256	[2][0][0] [ENTER] [5][0] [+] [6] [+]
2	**Combine 500 + 30 + 0.** Standard: CCSS.MATH.2.NBT.A.1	530	[5][0][0] [ENTER] [3][0] [+]
3	**Combine 600 + 0 + 9.** Standard: CCSS.MATH.2.NBT.A.1	609	[6][0][0] [ENTER] [9] [+]
4	**What is 100 more than 450?** Standard: CCSS.MATH.2.NBT.A.1	550	[4][5][0] [ENTER] [1][0][0] [+]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Ensure zeros in place values (e.g. 305 = 3 hundreds + 0 tens + 5 ones) are retained.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.